

SCHOOL OF ENGINEERING COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
B. TECH DEGREE SECOND SEMESTER SECOND INTERNAL EXAMINATION

CE A /CE B /ME B

23-200-0205A BASIC ELECTRONICS ENGINEERING (2023 scheme)

Time: 2 Hours

Max. Marks: 30

Course Outcomes:

On successful completion of this course the student will be able to:

CO1: Illustrate the concept of diode in rectifiers, filter circuits and wave shaping,

CO2: Interpret the functioning of oscillators and operational amplifiers.

CO3: Explain the principle of embedded systems and sensors

CO4: Summarise the functioning of a communication system and different modulation technologies

**Bloom's Taxonomy Levels (BL) : L1- Remember, L2-Understand, L3-Apply, L4-Analyze,
L5-Evaluate, L6 - Create**

PO - Programme Outcome

PART – A (Answer all questions.)

(5x 2 =10)

	Marks	BL	CO	PO
1. (a) What are the characteristics of an ideal op-amp?	2	2	2	1
(b) Define embedded systems. Give examples.	2	1	3	1
(c) Differentiate RISC and CISC processors.	2	1,2	3	1
(d) Differentiate input and output transducers with examples.	2	2	3	1
(e) Define a channel or medium.	2	2	4	1

Differentiate hardware and software channel with examples.

(P.T.O)

PART - B

(2 x 10 = 20)

	Marks	BL	CO	PO
2. (a) Explain the working of an Astable multivibrator using transistors with circuit diagram and necessary waveforms	7	2	2	1,5
(b). Explain the non-inverting configuration of an op-amp with circuit diagram	3	2	2	1
(OR)				
3. (a) Explain a basic communication system with block diagram.	5	1	4	1,5
(b) Explain the working of instrumentation and control systems with block diagrams.	5	2	3	1
(OR)				
4. (a) Mention the main elements of an embedded system with a schematic diagram and explain the core of an embedded system.	7	2	3	1,5
(b) Implement the EX-OR function using basic gates.	3	3	2	1
(OR)				
5. (a) Explain the working of a 7-segment LED display with suitable diagram and examples	6	2	3	1,5
(b) Explain the working of an analog transmitter with suitable block diagram.	4	1,2	4	1,5